K-Means Clustering to Identify Subscriber Segments in Cable Services

Customer Segmentation Case

Group 20.1
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A quick refresher...

Goal: Segment customers by service spending to optimize marketing offers

- Group MSO subscribers into distinct, data-driven customer clusters
- Identify high-value vs. budget-conscious subscribers
- Tailor personalized offers to each segment to maximize monthly revenue

Data & Methodology

📊 Data Overview

10,000 subscriber households (unit of analysis = household)

💭 Methodology

- Applied **K-means clustering** on video, internet, and phone spend
- Segmented households into distinct groups for targeted marketing
- Used demographics to profile segments and guide personalized offers

📺 Clustering Variables (for K-means):

- **Video** – Monthly spend on cable TV services (channels/packages)
- **Internet** – Monthly spend tied to speed and data plan usage
- **Phone** – Monthly spend on landline services

👪 Demographic Profiling Variables

Used to interpret clusters and design offers:

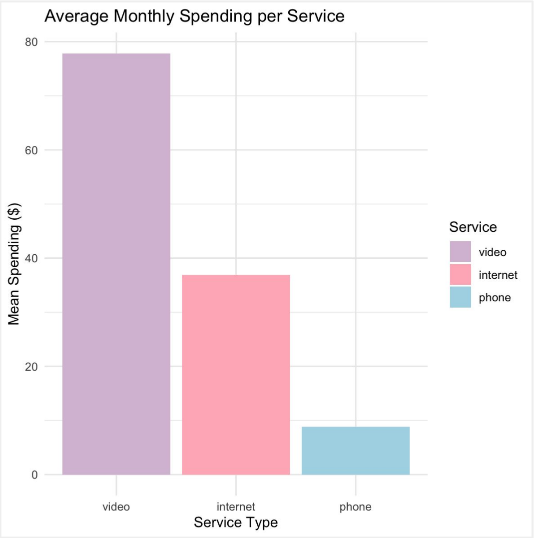
- Age of household head
- Household income
- Household size
- Marital status
- Number of children

Agenda

- Descriptive Statistics
- Standardization Decision
- Choosing the Best K-Means
- Cluster Profiles
- Marketing Strategy
- Data Limitations & Future Research

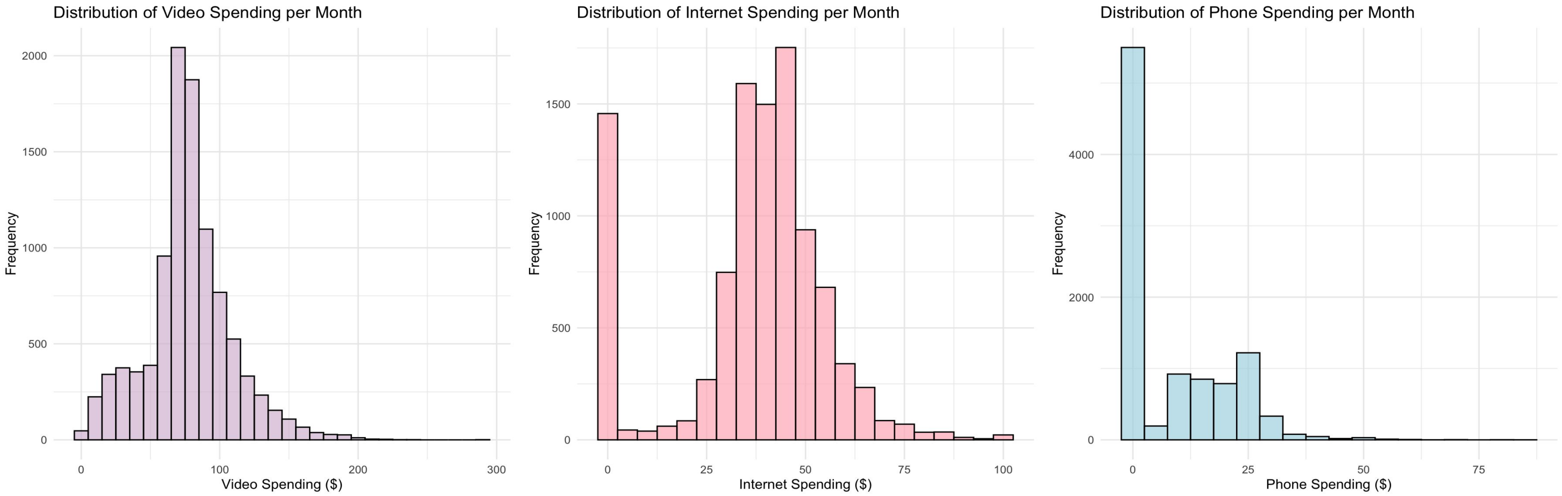
Highest Spend in Video, Lowest in Phone

Spending pattern	video	internet	phone
minimum	$0.39	$0	$0
1st quartile	$64.07	$31.30	$0
median	$76.17	$40.00	$0
mean	**$77.81**	**$36.89**	**$8.81**
3rd quartile	$92.72	$47.40	$17.75
maximum	**$293.72**	**$100**	**$84.95**



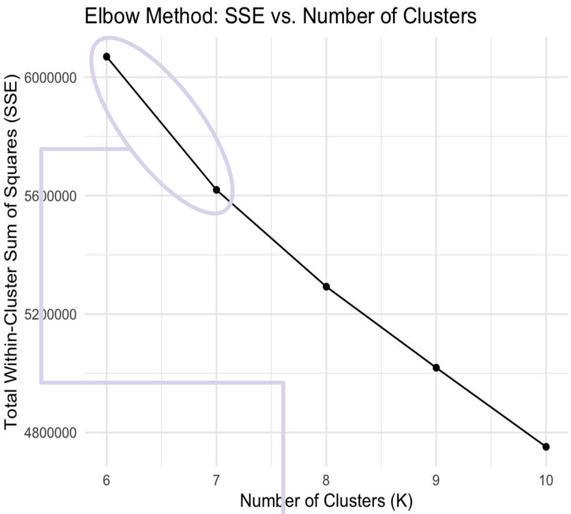
- **Video spending dominates**: Maximum video spend is nearly 3× higher than internet and over 3× higher than phone, suggesting video plans are the most expensive
- **Phone services underutilized**: Median and 1st quartile of phone spend are both zero, indicating many households don't pay for phone service
- **Internet spend stable**: Very little variation in internet spending, implying most households fall within a consistent price range

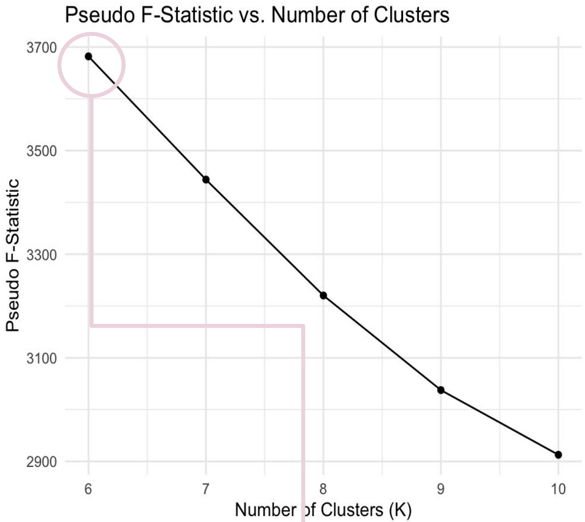
Household Spend Highlights **Shift from Phone to Internet**



- **Video & Phone**: Right-skewed distributions driven by a small group of high spenders
- **Internet**: Largely symmetric, suggesting spending is uniform across households
- **Phone usage**: Many households spend nothing on phone services, pointing to declining landline use and greater reliance on internet for communication and entertainment

Finding the Optimal Number of Clusters: **Considering SSE & Pseudo F**





K	SSE	Pseudo F
6	6,069,412	3,681.90
7	5,619,418 (-449,994)	3,444.05 (-237.85)
8	5,292,471 (-326,947)	3,220.32(-223.73)
9	5,018,888 (-273,583)	3,037.66 (-182.66)
10	4,751,659 (-267,229)	2912.75 (-124.91)

🔎**What We See**
Biggest drop in SSE from K=6 to K=7

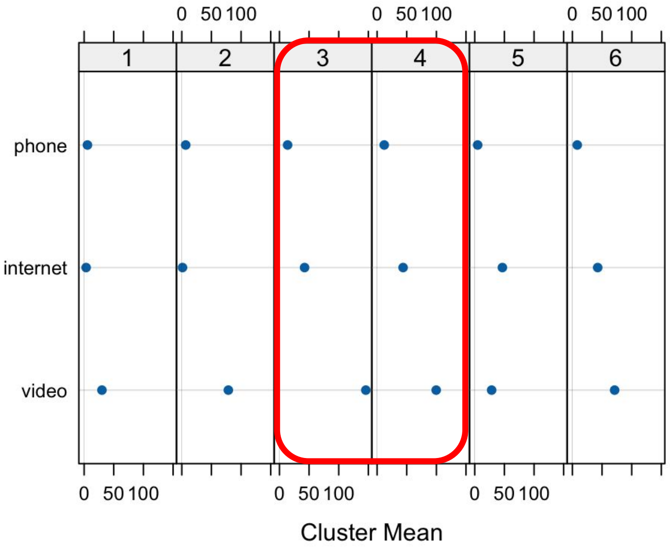
⭐**What This Means**
Adding the 7th cluster significantly reduces within-cluster variance

🔎**What We See**
Smallest Pseudo F at K=6

⭐**What This Means**
6 Clusters are optimal as it holds the highest Pseudo F score of all options

💡Let's test if **K=6** or **K=7** better optimizes customer segmentation for maximizing spend

K=6 Cluster Analysis: **Composition & Spending Patterns**



K	n	%	Video	Internet	Phone	Spending Key Trait
1	378	4%	$30.05	$3.39	$5.64	Lowest internet spend, low phone & video spend
2	1197	12%	$78.53	$1.24	$6.63	High video, low phone/internet spend
3	717	7%	$145.7	$42.40	$13.79	Highest video & phone spend, high internet spend
4	2344	24%	$99.68	$43.86	$12.07	2nd highest video, internet, & phone spend overall
5	1108	11%	$28.47	$46.65	$4.93	Lowest video & phone spend, highest internet spend
6	**4036**	**41%**	$71.22	$42.66	$8.22	Moderate spend overall

Initial Segmentation Concerns

🚩**Cluster 6 (41%) disproportionately large**, suggesting **potential need for finer segmentation** to uncover distinct spending patterns & optimize revenue strategies

🚩**Clusters 3 & 4 look very similar at first glance**, suggesting **redundancy & potential need for finer segmentation** to see clearer differentiation

K=6 Cluster Analysis: **Demographics Patterns**

K	n	Avg Age	Avg Income (1-9 Scale)	Household Size (1-9 people)	% Married	% w/ Children	Key Traits
1	378	57.8	3.67	2.29	44.7%	22%	Older, low income, small household, lower % married & child
2	1197	62.3	3.83	2.54	49.1%	24.3%	Older, low income, medium household & % married, low % child
3	717	53.4	5.87	3.26	67.9%	39.2%	Mid-age, high income, large household, higher % married & child
4	2344	52.1	5.24	3.02	59.2%	39.6%	Mid-age, high income, large household, higher % married & child
5	1108	46.2	4.31	2.49	44.1%	36.4%	Younger, middle income, small household, mid % married & child
6	4036	50.1	4.48	2.64	49.1%	34.1%	Younger, middle income, medium household, mid % married & child

Segmentation Concerns

🚩**Clusters 3 & 4 still seem very similar based on demographic patterns**, suggesting **need for finer segmentation** to see clearer differentiation

K=6 Cluster Analysis: **Profiling Issues**

K	Cluster Profile
1	Low-earning, older, independent individuals with simpler needs and lower tech reliance
2	Older, lower-earning households focused on security, stability, and essential needs
3	High-earning, middle-aged, family-oriented households with a strong emphasis on tech and (family) entertainment
4	High-earning, middle-aged, family-oriented households with a strong emphasis on tech and (family) entertainment
5	Middle-earning, young, tech-savvy individuals living in smaller households, focused on internet and personal tech
6	Middle-earning, middle-aged, family-oriented households balancing personal and family needs with moderate tech reliance

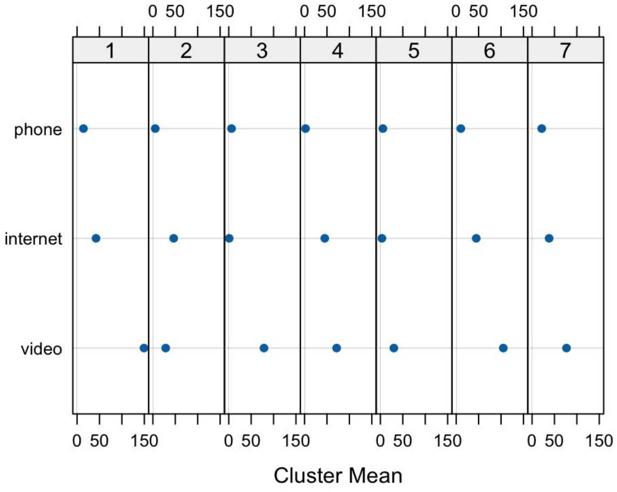
Segmentation Concerns & Conclusion:

🚩Clusters 3 & 4’s similar profiles limits their targeting effectiveness

🚩Cluster 6’s lack of distinct segmentation hinders targeted tactics

🔎**Analyzing K=7 may provide finer segmentation for more effective targeting**

K=7 Cluster Analysis: **Composition & Spending Patterns**



We believe that K=7 seems holds better-distributed composition than K = 6, so **ultimately went with 7 cluster segmentations**.

K	n	%	Video	Internet	Phone	Spending Key Trait
1	609	6%	$149.34	$42.27	$14.29	Highest Video Spending
2	1,113	11%	$28.52	$46.39	$5.25	Highest Internet Spending; Lowest Video Spending
3	1,179	12%	$78.7	$0.94	$6.26	Lowest Internet Spending
4	2,933	30%	$71.59	$45.08	$2.16	Lowest Phone Spending
5	378	4%	$30.10	$3.39	$5.63	Overall Bottom Spending
6	1,823	19%	$105.05	$44.65	$10.29	Overall Top Spending
7	1,745	18%	$76.86	$38.09	$21.62	Highest Phone Spending
	9,780	100	$77.96	$36.79	$8.88	

**high & low defined by cluster rankings & comparison with each variable's weighted average*

K=7 Cluster Analysis: **Demographic Patterns**

Cluster	Avg Age	Avg Income	Household Size	% Married	% w/ Children	Key Traits
1	53.5	5.85	3.25	68.5%	39.2%	Highest income; Larger-than-average households
2	46.5	4.32	2.49	44.7%	36.1%	Younger; Lower income, households size and children rate
3	62.4	3.81	2.54	48.8%	25.5%	Oldest; Lower income, households size, married and child presence
4	47.9	4.43	2.55	45.4%	33.8%	Second youngest out of all; Moderate income and marriage rate, Low child presence
5	57.8	3.69	2.29	45.8%	22%	Older;Lowest income; Smallest households; Almost no children
6	51.6	5.34	3.03	59%	39%	Middle-aged; 2nd highest income; Larger households; High children rate
7	55.1	4.76	2.94	59.3%	36.4%	Older; Mid-to-high income; Moderate marriage rate and child
Weighted avg	52.19	4.63	2.73	52.2%	34.4%	

**high, middle, & low defined by cluster rankings & comparison with each variable's weighted average*

K=7 Cluster Analysis: **Profiling**

K	Profile
1	**Premium-Loving, High-Spending Households across all services:** Middle-aged or older, high-income, larger households. Likely families who prioritize premium video and bundled services.
2	**Budget-Conscious Users, Younger, internet-reliant:** Middle-aged, price-sensitive customers, likely singles or couples who need affordable internet options. Career-focused household with 1 kid on average, prefer streaming over cable.
3	**Light-Use, Low-Spending Senior Customers / Potential Retirees:** Senior customers who still subscribe to traditional TV but have minimal interest in internet services.
4	**High-Volume Internet Users (Cord-Cutters) :** Working professionals or younger households who prioritize fast internet over cable TV.
5	**Very Low-Usage Customers:** Retired, fixed-income customers who use minimal services.
6	**Family-Oriented, High-Spending, Multi-Service Users, Entertainment-Focused:** Married households with kids who need balanced internet, phone, & TV services. Having highest overall spending, consistently exceeding average expenditures across all services. Within household age range broader, explaining broader needs.
7	**High-Spending, Traditional Households:** Older, more traditional mid-income customers who prioritize more traditional TV & phone services over newer tech trends

Defining Subscriber Segments by Business Value

High-Value Segments (Clusters 1, 6, 7)

- **Cluster 1:** Highest income, large households, high spend across all services → premium users
- **Cluster 6:** High income, large families, strong entertainment & bundle potential
- **Cluster 7:** Older, mid–high income, traditional households still spending on TV & phone

Traits:
💸 High spending
👩🏻‍💻 Strong retention
🆙 High upsell potential

Mid-Value Segments (Clusters 2, 4)

- **Cluster 2:** Younger, lower income, internet-reliant, value-driven bundle seekers
- **Cluster 4:** Younger, moderate income, cord-cutters but heavy internet users

Traits:
💸 Moderate spending
👩🏻‍💻 Growth potential via bundles & personalized offers

Low-Value Segments (Clusters 3, 5)

- **Cluster 3:** Oldest, low income, small households, minimal digital use
- **Cluster 5:** Retired, lowest income, very low spend across services

Traits:
💸 Low spending
👩🏻‍💻 Limited retention/upgrade potential

Tailoring Marketing Strategies by Subscriber Segments

High-Value Segments (Clusters 1, 6, 7)

Objective: Maximize revenue & long-term loyalty from premium, high-spending households

- Offer exclusive premium bundles (e.g., internet + video + family entertainment)
- Launch VIP loyalty perks (priority support, early access to new features)
- Promote upsell opportunities (higher-speed internet, add-on streaming, smart home services)

Traits:
💸 High spending
🧑🏻‍💻 Strong retention
🆙 High upsell potential

Mid-Value Segments (Clusters 2, 4)

Objective: Drive growth through engagement and upselling tailored to younger, internet-focused users

- Position affordable, flexible bundles to attract price-sensitive users
- Create personalized offers (e.g., streaming + high-speed internet combos)
- Target cord-cutters with enhanced internet plans and digital-first experiences

Traits:
💸 Moderate spending
🧑🏻‍💻 Growth potential via bundles & personalized offers

Low-Value Segments (Clusters 3, 5)

Objective: Maintain profitability while limiting investment in low-return households

- Provide simplified, low-cost plans to sustain essential services
- Focus on retention at minimal cost (basic loyalty offers, senior discounts)
- Avoid aggressive upselling; instead, streamline service touchpoints to reduce churn

Traits:
💸 Low spending
🧑🏻‍💻 Limited retention/upgrade potential

Data Limitations	Future Research
Segmentation currently relies only on **spending for video, internet, and phone**	Incorporate usage data (e.g., hours watched, streaming habits, device activity)
Spending data reveals *how much* customers pay, but **not *how they use* services**	Include demographic & lifestyle variables to refine customer profiles
Lack of behavioral & engagement metrics limits targeting, personalization, & churn prediction	Explore longitudinal data to track changes in behavior and optimize offers over time

More Important Data to Collect (Internet Usage, Customer Loyalty, Household & Device Data, Cost of Service)

Category	Additional Variables	Purpose
Business profitability & cost	Customer Loyalty & Churn Risk	**Years as a customer** → Reward long-term users before they consider leaving. **Payment & service complaints** → Identify churn risks early. **Customer service calls** → Offer personalized retention deals to frustrated users.
	Cost Data	**Service cost per customer** → Helps distinguish high-revenue vs. high-cost customers. **Profitability per segment** → Prevents us from prioritizing expensive-to-serve customers. **Cost-to-revenue ratio** → Identifies which segments generate the most profit, not just revenue.
User behavior & usage patterns	Internet Usage	**Total data used per month** → Find heavy users for speed upgrades. **Streaming service usage** →Target cord-cutters with streaming bundles. **Peak usage times** → Offer late-night plans to night streamers.
	Household & Device Data	**Years as a customer** → Reward long-term users before they consider leaving. **Payment & service complaints** → Identify churn risks early. **Customer service calls** → Offer personalized retention deals to frustrated users.

Better Data = More Revenue & Lower Churn

✅ **More targeted promotions increase revenue**

- Sell gigabit internet to heavy data users.
- Offer premium sports channels to sports fans.
- Push mobile-friendly plans to streaming-focused users.

✅ **Identifying churn risks early helps prevent cancellations**

- If a customer rarely uses cable but streams a lot, offer a streaming-first package.
- If a customer calls support often, offer a discount to improve retention.
- If a long-term customer has never upgraded, provide a loyalty deal before they switch providers.

📢 **Next Steps:** Start collecting this data and **test how it impacts revenue and retention.**

Questions or Feedback?

Let’s discuss.